

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAPE504D - AUTOMOBILE ENGINEERING

Time : 1.5 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The basic automobile structure consists of the suspension system. axles, wheels and
(a) steering (b) brakes (c) frame (d) lights
The purpose of gear box in an automobile is to
(a) vary speed (b) vary torque (c) provide permanent speed reduction
(d) to disconnect the road wheels from the engine when desired
The most effective section against bending is
(a) rectangular bar (b) round hollow tube (c) round bar (d) square hollow section
- 4 The rigid suspension is beneficial when it is desired to
(a) Reduce the unsprung mass,
(b) Improve tyre-to-ground contact characteristics
(c) Have more flexibility in design
(d) Have large suspension stroke to take care of large changes in load
Hard steering is due to
(a) Low tyre pressure (b) High speed of vehicle (c) Worn out bearing of front wheels
(d) Stub axle bending
Automobile gears are generally made of
(a) Cast iron (b) Mild steel (c) Wrought iron (d) Carbon crone manganese steel
- 7 The purpose of torque converter in automobile is
(a) Automatically multiplying engine torque (b) Automatically multiplying engine speed
(c) Automatically control the speed of engine (d) Automatically multiplying vehicle torque

- 8 The hole or slot in the rim which accommodates the valve for tyre inflation is called-
 (a) Well (b) Flange (c) Valve aperture (d) None of these
- 9 In heavier vehicles, tubeless tyres cannot be used because
 (a) Rims are flat based and they are airtight (b) Rims are flat based and they are not airtight
 (c) Rims are well shaped and airtightness is present
 (d) Rims are well shaped but no airtightness is present
- 10 Tandem master cylinder consists of
 (a) One cylinder and one reservoir (b) Two cylinders and one reservoir
 (c) One cylinder and two reservoirs (d) Two cylinders and two reservoirs
- 11 Mechanical efficiency of an automobile engine usually varies in the range
 (a) 50-60% (b) 60-70% (c) 70-90% (d) more than 90%
- 12 The most accurate dynamometer is the
 (a) prony brake type (b) hydraulic type (c) swinging field type
 (d) eddy current type
- Q2. Solve the following.
- A) What are the different loads acting on chassis and explain briefly? 6
- B) Justify the need of a gear box with suitable sketch. 6
- Q3. Solve the following.
- A) Explain a typical power steering system. 6
- B) What are the objectives and components of suspension system? 6
- Q4. Solve Any Two of the following.
- A) Explain the purpose and working of differential unit. 6
- B) Explain the working principle of synchromesh gear box with neat sketch. 6
- C) Explain the working of universal joint with neat sketch. 6
- Q5. Solve Any Two of the following.
- A) Explain the Pneumatic and Hydraulic Braking Systems in a Automobile with suitable schematic diagram. 6
- B) Define caster and camber with suitable schematic diagram. 6
- C) Define and explain the following front-wheel alignment factors: • King-pin inclination • Toe-in • Toe-out 6
- Q6. Solve Any Two of the following.
- A) With a neat sketch, explain the construction of a lead acid battery. 6
- B) Explain the principle of operations, constructions and working of starting Motor. 6
- C) Write in details about the engine tune up methodology. 6

*** End ***